MIL-T-82374A(TD)
4 January 1978
SUPERSEDING
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MILITARY SPECIFICATION

TRAINING AID, CUTAWAY,

INTERNAL COMBUSTION ENGINE

This specification has been approved by the Naval Training Equipment Center, Department of the Navy.

1. SCOPE

1.1 This specification covers the requirements for the modifications required to convert a 6 cylinder internal combustion engine into a training aid.

1.1.1 This item has been designated by the Naval Training Equipment Center as Device 17F30 for Navy procurement (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation; Methods of

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-129	Marking for shipment and storage
MIL-STD-794	Parts and equipment; Procedure for packaging and packing of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Training Equipment Center, Orlando, FL 32813, Attn: Code N423, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6910

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DRAWINGS

Naval Training Equipment Center

L. E. O'Neil, Nos. 2300 through 2327	Cutaway, Internal Combustion Engine, Device 6A2

PHOTOGRAPHS

Naval Training Equipment Center

Numbers: BXA 9869-2-67 BXA 9870-2-67 BXA 9871-2-67	Photographs, Cutaway, Internal Combustion Engine, Device 6A2

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Materials. - Materials for the training aid shall be as specified on the drawings of 2.1. Materials not specified on the drawings of 2.1 shall be suitable for the application and satisfy the requirements under 3.

3.2 Design. - A six cylinder, four cycle gasoline engine shall be modified to serve as a training aid. The engine shall be sectionalized so that the principles of the operation can be clearly demonstrated. The sectionalized engine shall be rotated at low speed by an electric motor.

3.2.1 The engine shall be mounted on a support stand with locking casters. The stand shall support and contain the operational electric system.

3.3 Reliability. - Reliability of operation shall be a prime consideration in the design and manufacture of the training aid. The design shall be such as to result in reduced requirements for adjustments and alinement, and low frequency of failure, while providing for simplified maintenance. In addition, the training aid shall be constructed so no fixed part, assemblie or subassemblies shall bind or become undesirabily free. Degradation shall not be caused in the mechanical performance stipulated for the device because of shipment, normal handling or storage.

3.4 Construction. - Construction shall be in accordance with the drawings listed under 2.1. These drawings are not guaranteed for accuracy and completeness, but are considered adequate for use by a contractor, with experienc in the sectionalizing of engines.

3.5 Performance. - The training aid shall withstand 20 hours per week use in a classroom for 10 years with routine maintenance only.

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3.6 Details of components.- The training aid shall consist of the following major items:

(a) Sectionalized engine

(b) Operational electric system

(c) Support stand

(d) Dust cover

(e) Spare parts

3.6.1 Engine.- The engine to be sectionalized shall be a six cylinder, four cycle gasoline engine, Chevrolet truck engine, or equal. A used engine with all components intact may be processed.

3.6.1.1 The engine shall be sectionalized in accordance with the drawings under 2.1 and as specified under 3.6.1.

3.6.1.2 Sliding surfaces.- All sliding surfaces in the engine, such as cylinder-piston, connecting rod and main bearings shall be prepared by suitable means to reduce friction to a minimum. Clearances between sliding bearings shall be increased, but not made excessively loose. Piston rings in the closed cylinders shall be removed and teflon pads shall be substituted in upper and lower piston groove only. All inner cylinder surfaces shall receive a coating of chromium. All spring pressures shall be reduced for easy operation.

3.6.1.3 Shields.- Sectionalized areas, which represent a hazard due to the moving parts shall be covered by clear acrylic covers. These covers shall be shaped to fit their supporting contours without strain.

3.6.1.4 Color coding.- All passages, manifolds, tubes and pipes containing fluid shall be color coded in accordance with Table I. All passages shall be sectionalized part of their length. Cut open passages shall have solid color on the inside. Closed pipes shall show transverse stripes on the outside, spaced at intervals to identify the content. All paint shall have semi-gloss finish.

3.6.1.4.1 The direction of the flow shall be indicated by arrows in contrasting colors spaced at suitable intervals. Inlet and outlet, where appropriate, shall be shown by suitable arrows with nomenclature.

3.6.1.4.2 A nomenclature plate, made from 1/8 inch aluminum alloy shall be fastened to the support stand. The nomenclature plate shall identify colors by sample and legend. See also Table I. The lettering shall be readable at eight (8) feet distance. Colors, codes and lettering shall be protected by a clear satin-finish lacquer or equivalent plastic sheet.

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3.6.1.4.3 All colors shall be in accordance with FED-STD-595. All exterior surfaces of the engine shall be painted, semigloss finish, FED-STD-595 No. 26314 gray. The interior surfaces of crankcase and pan shall be No. 26622 gray. All cut metal surfaces shall be No. 27886 white.

3.6.2 Operational electrical system.- The operational electrical system supplies current and control for illumination, timing, spotlight pointer and motor for engine rotation. The system operates on 24 Volt AC supplied by a transformer from a power source of 115 Volts, 60 Hertz. All metal parts of the training aid shall be grounded through the ground-line of the electrical supply. A control and storage box shall house the major elements of the system.

3.6.2.1 Illumination.- Interior sections of the engine, such as crankcase, waterjackets and manifolds shall be illuminated by C-8 type 24V AC white bulbs. These bulbs shall be arranged to be out of direct view. The current shall be supplied through the operational electrical system described under 3.6.2.

3.6.2.2 Timing.- The explosion stroke in each cylinder shall be simulated by the flashing of a T-4 type red 24V AC bulb. In the cut-open cylinders it shall be located in the spark plug. In the closed cylinders the bulb shall be on the outside, close to the spark plug. The timing shall be controlled by the engine timing system.

3.6.2.3 Control and storage box.- The control and storage box shall be an all metal container with door, mounted on the support stand. A 20 foot three conductor cable with plug shall be provided for connection to the power source. A 10 foot flexible cable shall extend from the box to a handgrip which carries a spotlight pointer with switch and the control switch for the engine drive motor. The spotlight pointer shall produce a high intensity beam of light making a 1-1/2 inch spot at four feet distance.

3.6.2.3.1 On the outside of the control box shall be mounted: one main power control switch with an on and off light, one switch for engine illumination circuit, one switch for engine timing circuit.

3.6.2.3.2 The inside of the control box shall house: one 115 to 24 volts step-down transformer, main fuse, one fuse each for motor, spot-light, illumination and timing circuit. Storage space for the two flexible cables under 3.6.2.3 shall be provided. Storage space for spare parts shall be provided.

3.6.2.3.3 Engine drive motor.- The engine drive motor operating from 24V AC, shall be installed in the support stand and shall have rigid conduits to the control box. The drive motor shall be equipped with suitable reduction gearing and a belt drive to the engine. The section-alized engine shall rotate at approximately six (6) revolutions per minute.

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3.6.2.3.3.4 Safety clutch.- An adjustable safety clutch shall be provided in the engine drive system.

3.6.3 Support stand.- A support stand shall be provided in accordance with the drawings under 2.1.

3.6.3.1 The support stand shall have four casters, each with lock for wheel and swivel. The casters shall have a hard phenolic tread and a minimum load rating of 1000 pounds each.

3.6.3.2 The support stand shall be painted semi-gloss dark gray corresponding to FED-STD-595 No. 26152.

3.6.4 Dust cover.- A dust cover of gray, 12 ounce vinyl coated nylon equal to the tensile, tear and flame resistance characteristics of Armor-Tite shall be provided to protect the training aid when not in use. The covering material and material used in the construction of the dust cover shall be compatible with all finishes of the training aid.

3.6.5 Spare parts.- Three sets of spares shall be provided. Each set consists of all the bulbs and fuses used in the training aid. These spare parts shall be packed in properly labeled plastic boxes and stored in the control and storage box (see 3.6.2.3.2).

3.7 Finish and color.- Finish and color shall be in accordance with the drawings of 2.1 and Table 1 below. In case of a conflict between the drawings and Table 1, Table 1 shall supersede.

TABLE I

Usage	Color	FED-STD-595 Number
Exterior engine	Medium gray	26314
Interior engine	Light gray	26622
Cut sections	White	27886
Fuel	Light yellow	23793
Lubrication oil	Dark yellow	23538
Coolant	Medium green	24260
Intake air	Light blue	25488
Exhaust	Orange	22510

3.8 Nameplates and product markings.- Nameplates and product markings will be specified by the procuring activity and shall be permanently affixed to the training aid for identification.

3.9 Workmanship.- The training aid shall be constructed and finished with particular attention given to neatness, sturdiness, correctness of detail, freedom from sharp edges and burrs, ease of operation and clarity, sharpness and accuracy of lettering.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections.- Inspections to be performed are classified as follows:

(a) In-process inspection

(b) Quality conformance inspection

4.2.1 In-process inspection.- The in-process inspection shall include such visual and mechanical examinations and testing of materials, sub-assemblies, parts and accessories (including purchased standard parts) during the manufacturing process as may be necessary to assure conformance to all the requirements of this specification.

4.2.2 Quality conformance inspection.- The quality conformance inspection shall consist of the examinations and tests of 4.3 and 4.4.

4.3 Examinations.- Examinations shall be performed to verify conformance to the following requirements:

(a) Materials

(b) Construction

(c) Details of components

(d) Color and finish

(e) Nameplates and product markings

(f) Workmanship

(g) Preparation for delivery

4.4 Tests.- Each training aid shall be subjected to functional tests covering all possible demonstrations required by the design requirements. A detailed outline of these tests will be furnished by the Inspection and Test Check List, a part of the Government-furnished property, specified by the procuring activity.

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5. PACKAGING

5.1 Preservation-packaging.- Preservation-packaging shall be in accordance with MIL-P-116, Method III.

5.2 Packing.- The training aid shall be packed in accordance with MIL-STD-794, level to be specified by the procuring agency.

5.3 Marking.- Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The training aid will be used to familiarize students with the theory and principles of operation, maintenance and repair of an internal combustion engine.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification

(b) Procuring activity designation (device number or other) (see 1.1.1).

(c) Selection of applicable levels of preservation, packaging and packing (see 5.1 and 5.2).

(d) Nameplate data.

6.3 Definitions.- Except as specified in 6.3.1 definitions will be furnished upon written request to the procuring activity.

6.3.1 Training aid.- A training device designed to demonstrate or illustrate a concept, or to portray the functional characteristics of an end item without use of the actual working medium as to motivating force. Usually instructor operated and may also be used by the student for self-instruction.

CUSTODIAN:
Navy-TD

PREPARING ACTIVITY:
Navy-TD
Project No. 6910-N-375

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-T-82374A(TD), TRAINING AID, CUTAWAY, INTERNAL COMBUSTION ENGINE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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